TASK SPECIFICATION: P7-02 PHASE P7 — NOTIFICATIONS, GOVERNANCE & TELEMETRY ANALYTICS

Event-Driven Notification Bus Dispatcher

Git Branch: `feature/P7-02-notification-events` · Target Pull Request: feature/P7-02-notification-events

Core Objective & Technical Scope:
Decoupled Spring application event listeners automatically triggering notifications for parking reservations, payments, and tutoring sessions.

Priority Level	High	Complexity Rating	`M`
Execution Status	Backlog 📋	Target PR / Commit	`feature/P7-02-notification-events`
Upstream Dependencies	P7-01, P4-02, P5-03, P6-02, P6-04		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Define domain events (BookingCreatedEvent, PaymentProcessedEvent, SessionBookedEvent).
2. Implement asynchronous @TransactionalEventListener handlers in notification module.
3. Format bilingual notification templates with action links.
4. Write integration tests verifying end-to-end event propagation.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `Domain modules: event publishers`
- `notification/application/listeners/*`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- Every major business transaction produces an in-app notification reliably.
- Event failures do not break the underlying business transaction.

VERIFICATION & TEST COVERAGE STRATEGY

- NotificationEventListenerIntegrationTest